

Ziru Wei

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Keywords: Human-Computer Interaction; Human-Robot Interaction; Context-Aware Interaction

RESEARCH INTERESTS

My research explores next-generation AI interfaces designed to augment humans undistruptively in everyday physical environments. I develop medium-agnostic and context-adaptive systems spanning modalities from extended reality to robotic interfaces, and investigate how humans perceive them.

EDUCATION

The ATLAS Institute, University of Colorado Boulder

Starting Fall 2026

- PhD in Creative Technology and Design
- Advisor: Prof. Ryo Suzuki

Carnegie Mellon University

Graduated May 2026

- Master of Science in Computational Design (research-based)
- Advisor: Prof. Alexandra Ion
- Thesis Project: *Toward Interacting with Proactive Intelligence in Everyday Physical Environments*

Soochow University

Graduated Jun 2024

- Bachelor of Architecture

RESEARCH WORKS

[R1] (In submission) While You're Still Busy: Context-Aware Approach Optimization for Proactive Mobile Robots

Ziru Wei, Di Wen, Leyuan Chen, Hassan Shahzad, Alexandra Ion

[R2] (In submission) "Let Me Lend You a Hand": Understanding Contextual Perceptions of Physical Proactivity in Small-Scale Personal Assistance Robots

Ziru Wei, Violet Yinuo Han, Tanvi Handoo, Alexandra Ion

[R3] Objects with Arms: How Task-Relevant Embodiment Shapes Perception in Physical Collaboration

Violet Yinuo Han, Ziru Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2026 (full paper accepted)

[R4] Embodied Generative Taskscape: Re-Connect Cultural-Ecological Perception in 5-Animal Play

Ziru Wei, Jimmy Cheng. *50th Annual Meeting of the Society for Social Studies of Science (4S)*, 2025 (abstract accepted)

[R5] On-site Holographic Building Construction: A Case Study of Aurora

Sijie Liu, Ziru Wei, Sining Wang. *Proceedings of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, 2022 (full paper accepted, ~30% acceptance, peer-reviewed, top-tier conference in computational design)

RESEARCH EXPERIENCES

Master's Student | Interactive Structures Lab, Carnegie Mellon University

Mar 2025 - Present

- Advisor: Prof. Alexandra Ion
- Developed software and hardware systems for proactive robotic interfaces that blended into everyday environments, and understood human perceptions of them (See R1, R2, R3).

Research Intern | WHY Research Lab, School of Architecture, Carnegie Mellon University Aug 2024 - Jan 2025

- Advisor: Prof. Daragh Byrne
- Replicated an e-waste scanner by disassembling and re-soldering disk drives, integrating a Raspberry Pi controller; built the WasteStation database in Notion to map components and their potential reuse applications

Research Intern | Humachine Lab, Architecture Department, Soochow University

May 2021 - Aug 2022

- Advisor: Prof. Sining Wang
- Designed MR workflows and four on-site collaboration methods for nonlinear façade assembly with limited devices and a small construction team; documented the design-to-construction process and contributed to the literature review part in paper writing (See R5)

AWARDS AND HONORS

Computational Design Commendation, Carnegie Mellon University	2025
Computational Design Commendation, Carnegie Mellon University	2024
Merit Scholarship, Carnegie Mellon University	2024
Outstanding Undergraduate Thesis (Top 1%), Soochow University	2024
Excellence Award, Shanghai Youth Architectural Design Competition	2023
Innovation & Academic Excellence Scholarship, Soochow University	2020 - 2022
Overall Excellence Award Winner, Solar Decathlon China	2022
First Prize (Top 2%) in “Zijin Award” of Architectural Design Contest	2022
METTLER TOLEDO Scholarship (Top 2%)	2019

MENTORING

Tanvi Handoo , Undergraduate Student in Computer Science, Carnegie Mellon University	May 2025 - Sept 2025
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ACTIVITIES

Course Guest Reviewer, Carnegie Mellon University	Jan 2025 - May 2025
Student Volunteer, des[AI]gn conference 2024, American Institute of Architecture Students	Oct 2024

SKILLS

Technical

- Hardware: Arduino, Raspberry Pi
- Programming: Python, PyTorch, Rstudio, C#, Pascal, HTML, CSS, JavaScript

Design & Production

- Software: Unity, Rhino, Grasshopper with GHPython, Blender, AutoCAD, Adobe Creative Suite, Figma, Procreate
- Fabrication: 3D printing, Soldering and electronic wiring, Welding (basic), Woodworking

Languages

- English (Fluent), Mandarin (Native), Portuguese (Beginner)